Data Mining Project - Customer Behavior Analytics & Predictive Insights from E-Commerce Data

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Submitted by:

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1. Introduction & Problem Statement:

Modern e-commerce platforms generate vast volumes of transactional data on a daily basis. While this data contains valuable information, extracting actionable insights from raw transaction logs remains a significant challenge for businesses. This project addresses three key business problems:

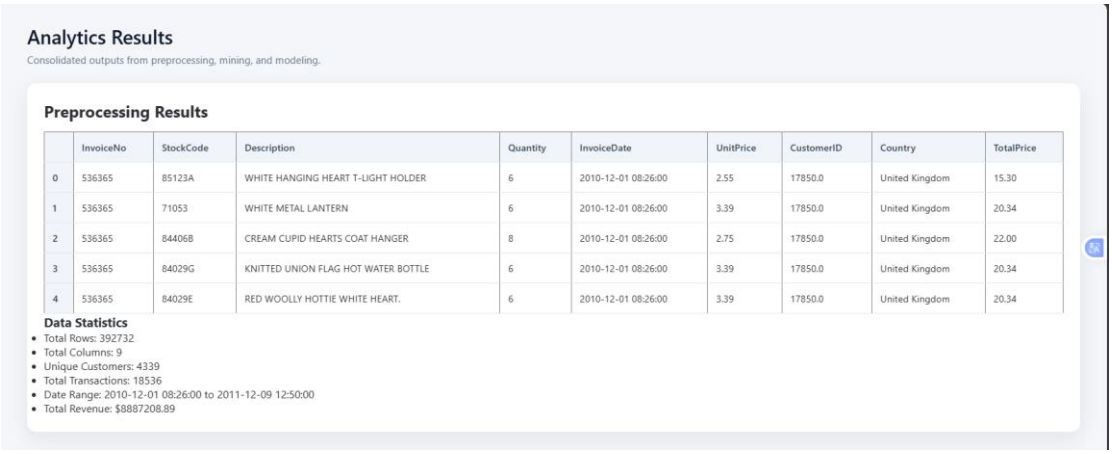
1. **Unknown Associations:** Identifying products that are frequently purchased together.
2. **Customer Heterogeneity:** Segmenting customers into meaningful groups based on purchasing behavior.
3. **High-Value Prediction:** Predicting which customers are likely to become high spenders.

To address these challenges, the **Online Retail Dataset** from the **UCI Machine Learning Repository** was used. The project applies **Exploratory Data Analysis (EDA)**, **Association Rule Mining (Apriori)**, **Customer Segmentation (K-Means Clustering)**, and **Classification Models** to derive meaningful business insights.

2. Data Preprocessing:

The dataset consisted of raw transactional records, which required extensive preprocessing to ensure analytical accuracy and reliability. The following steps were implemented in Python:

1. **Handling Missing Values:**
   Records with missing CustomerID values were removed, as customer-centric analysis (such as RFM) requires unique customer identifiers.
2. **Cleaning Invoices:**
   Transactions with InvoiceNo starting with 'C' (cancellations) were excluded to prevent distortion in revenue and frequency calculations.
3. **Data Validity Checks:**
   Rows containing negative values for Quantity or UnitPrice were filtered out, as these entries are not valid retail transactions.
4. **Feature Engineering:**
   A new feature, TotalSpent, was created by multiplying Quantity with UnitPrice.Additionally, InvoiceDate was converted into a datetime format to support temporal and recency-based analysis.



Analytics Results

Consolidated outputs from preprocessing, mining, and modeling.

Preprocessing Results

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country	TotalPrice
0	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	2010-12-01 08:26:00	2.55	17850.0	United Kingdom	15.30
1	536365	71053	WHITE METAL LANTERN	6	2010-12-01 08:26:00	3.39	17850.0	United Kingdom	20.34
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	2010-12-01 08:26:00	2.75	17850.0	United Kingdom	22.00
3	536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6	2010-12-01 08:26:00	3.39	17850.0	United Kingdom	20.34
4	536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	2010-12-01 08:26:00	3.39	17850.0	United Kingdom	20.34

Data Statistics

- Total Rows: 392732
- Total Columns: 9
- Unique Customers: 4339
- Total Transactions: 18536
- Date Range: 2010-12-01 08:26:00 to 2011-12-09 12:50:00
- Total Revenue: $8887208.89

3. Exploratory Data Analysis (EDA):

3.1 Top Selling Products

A bar chart was generated to visualize the **top 10 products by quantity sold**.

1. **Observation:** The most frequently sold items include *"World War 2 Gliders"* and *"White Hanging Heart T-Light Holder"*.
2. **Insight:** The product portfolio primarily consists of decorative and gift items, indicating that the store functions more as a gift shop rather than a seller of high-value electronics or furniture.

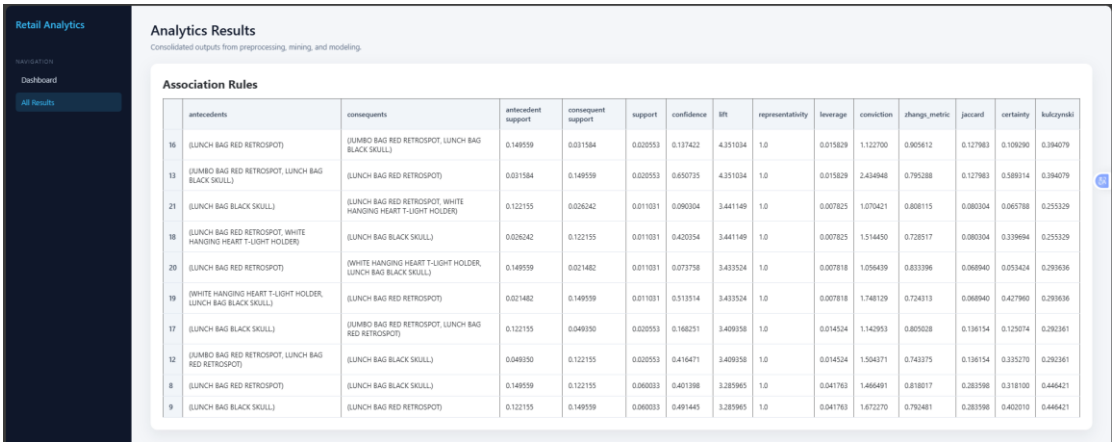
3.2 Monthly Revenue Trend

A line chart was used to analyze monthly revenue trends from late

1. **Observation:** While sales remain relatively stable throughout the year, a significant spike occurs in **November**.
2. **Insight:** This spike corresponds to **Christmas shopping**, suggesting strong seasonality. Businesses should increase inventory and marketing efforts during October to maximize sales during this peak period.

4. Association Rule Mining (Apriori):

Association Rule Mining was performed using the **Apriori algorithm** to conduct **Market Basket Analysis** on the **France subset** of the dataset. This analysis identifies product combinations that are frequently purchased together.



Retail Analytics

NAVIGATION

Dashboard

All Results

Analytics Results

Consolidated outputs from preprocessing, mining, and modeling.

Association Rules

	antecedents	consequents	antecedent support	consequent support	support	confidence	lift	representativity	leverage	conviction	zhangs_metric	jaccard	certainty	kulczynski
16	(LUNCH BAG RED RETROSPOT)	(JUMBO BAG RED RETROSPOT, LUNCH BAG BLACK SKULL)	0.149559	0.031584	0.020553	0.137422	4.351034	1.0	0.015829	1.122700	0.905612	0.127983	0.109290	0.394079
13	(JUMBO BAG RED RETROSPOT, LUNCH BAG BLACK SKULL)	(LUNCH BAG RED RETROSPOT)	0.031584	0.149559	0.020553	0.650735	4.351034	1.0	0.015829	2.434948	0.795288	0.127983	0.589314	0.394079
21	(LUNCH BAG BLACK SKULL)	(LUNCH BAG RED RETROSPOT, WHITE HANGING HEART T-LIGHT HOLDER)	0.122155	0.026242	0.011031	0.090304	3.441149	1.0	0.007825	1.070421	0.808115	0.080304	0.065788	0.255329
18	(LUNCH BAG RED RETROSPOT, WHITE HANGING HEART T-LIGHT HOLDER)	(LUNCH BAG BLACK SKULL)	0.026242	0.122155	0.011031	0.420354	3.441149	1.0	0.007825	1.514450	0.728517	0.080304	0.339694	0.255329
20	(LUNCH BAG RED RETROSPOT)	(WHITE HANGING HEART T-LIGHT HOLDER, LUNCH BAG BLACK SKULL)	0.149559	0.021482	0.011031	0.073758	3.433524	1.0	0.007818	1.056439	0.833396	0.068940	0.053424	0.293636
19	(WHITE HANGING HEART T-LIGHT HOLDER, LUNCH BAG BLACK SKULL)	(LUNCH BAG RED RETROSPOT)	0.021482	0.149559	0.011031	0.513514	3.433524	1.0	0.007818	1.748129	0.724313	0.068940	0.427960	0.293636
17	(LUNCH BAG BLACK SKULL)	(JUMBO BAG RED RETROSPOT, LUNCH BAG RED RETROSPOT)	0.122155	0.049350	0.020553	0.168251	3.409358	1.0	0.014524	1.142953	0.805028	0.136154	0.125074	0.292361
12	(JUMBO BAG RED RETROSPOT, LUNCH BAG RED RETROSPOT)	(LUNCH BAG BLACK SKULL)	0.049350	0.122155	0.020553	0.416471	3.409358	1.0	0.014524	1.504371	0.743375	0.136154	0.335270	0.292361
8	(LUNCH BAG RED RETROSPOT)	(LUNCH BAG BLACK SKULL)	0.149559	0.122155	0.060033	0.401398	3.285965	1.0	0.041763	1.466491	0.818017	0.283598	0.318100	0.446421
9	(LUNCH BAG BLACK SKULL)	(LUNCH BAG RED RETROSPOT)	0.122155	0.149559	0.060033	0.491445	3.285965	1.0	0.041763	1.672270	0.792481	0.283598	0.402010	0.446421

5. Customer Segmentation (Clustering):

Customer segmentation was performed using the **RFM (Recency, Frequency, Monetary)** framework, followed by **K-Means clustering** with **k = 4**.

Cluster Profiles

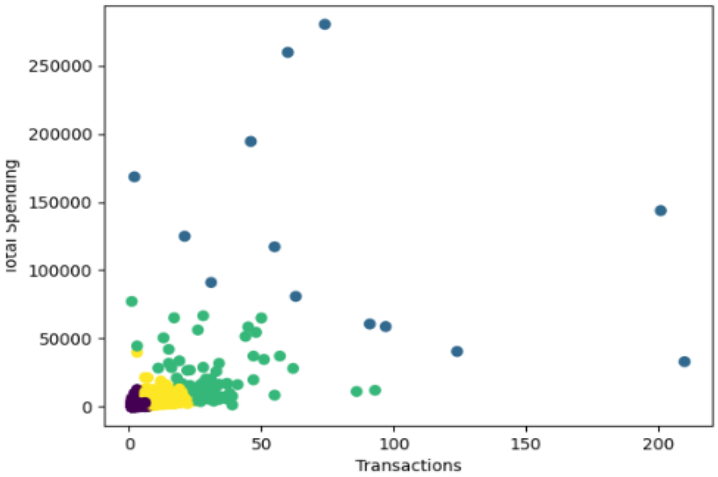
1. **Cluster 0 – Loyal Customers:**
   Moderate recency and frequency. These customers make consistent purchases over time.
2. **Cluster 1 – At-Risk Customers:**
   High recency (long time since last purchase) and low frequency. These customers are likely to churn.
3. **Cluster 2 – VIP Customers:**
   Very high monetary value and high frequency. Although small in number, this group contributes disproportionately to total revenue.

Recommended Actions:

1. **VIP Customers:** Provide exclusive offers and early access to new products.
2. **At-Risk Customers:** Launch re-engagement campaigns using personalized discounts or coupons.

Customer Segmentation

Silhouette Score: 0.6982
Calinski-Harabasz Score: 3123.1414
Davies-Bouldin Score: 0.8275



6. Classification Models:

A supervised learning task was formulated to predict whether a customer belongs to the **"High Value"** category (top 25% based on monetary value) using behavioral features such as **recency and frequency**.

Model Performance

Both Logistic Regression and KNN models demonstrated strong predictive performance. The confusion matrix shows a **very low false positive rate**, ensuring that premium marketing efforts are targeted only at genuinely high-value customers.

Classification Results

Accuracy: 0.8490783410138248

	precision	recall	f1-score	support
0	0.81	0.89	0.85	410
1	0.89	0.81	0.85	458
accuracy			0.85	868
macro avg	0.85	0.85	0.85	868
weighted avg	0.85	0.85	0.85	868

7. Conclusion & Future Work

This project successfully transformed raw transactional data into actionable business insights. The analysis revealed that the French market is driven by **set-based purchases**, and a clearly identifiable **VIP customer segment** plays a critical role in revenue generation.

For future work, the system can be extended by implementing a **Collaborative Filtering Recommendation System** to provide personalized product recommendations and further enhance customer experience.

8. Teamwork and Collaboration:

1. Fatima Liaquat: Data Preprocessing & EDA.
2. Marium Imran: Clustering & Classification Models
3. Sanawar: Association Rule Mining & Report Compilation.